

FREDRIKA, Sweden

WMO: 022630

Lat: 64.0744N

Lon: 18.3619E

Elev: 1073

StdP: 14.14

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.1	-7.2	-17.5	2.2	-13.1	-11.1	3.1	-7.0	22.1	27.3	15.1	25.7	1.6	350	0.649

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	76.8	60.5	73.2	59.7	69.7	58.1	63.7	71.4	61.8	69.5	59.9	66.9	5.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.9	83.0	66.1	58.8	77.0	64.1	56.9	71.9	62.0	29.5	71.5	28.1	69.5	26.8	67.1	70.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.5	11.7	10.1	-20.9	80.8	5.7	3.9	-25.0	83.6	-28.4	85.9	-31.6	88.1	-35.7	90.9		
(5)			-20.9	66.1	5.4	2.3	-24.8	67.7	-28.0	69.1	-31.1	70.4	-35.0	72.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.4	16.0	16.9	24.6	34.8	45.1	53.3	59.0	55.6	46.8	36.2	26.5	20.5		
	DBStd	16.92	11.50	11.59	8.92	6.37	7.12	6.08	5.49	5.44	5.64	6.79	8.63	10.94		
	HDD50	5631	1053	927	788	457	188	33	2	12	125	428	704	914		
	HDD65	10460	1518	1347	1253	907	617	355	199	294	545	892	1154	1379		
	CDD50	666	0	0	0	0	36	131	281	187	30	1	0	0		
	CDD65	21	0	0	0	0	0	3	14	4	0	0	0	0		
	CDH74	246	0	0	0	0	6	58	148	34	0	0	0	0		
	CDH80	25	0	0	0	0	0	4	20	1	0	0	0	0		
	WSAvg	4.7	4.2	4.3	5.2	5.1	5.3	5.3	4.6	4.3	4.5	4.7	4.3	4.1		
	PrecAvg	23.5	1.5	1.1	1.1	1.2	1.6	2.4	3.6	3.2	2.3	2.0	1.9	1.7		
(15) Precipitation	PrecMax	32.5	2.7	2.7	2.6	2.2	2.8	4.1	8.6	6.7	5.5	5.2	3.8	3.4		
	PrecMin	17.2	0.1	0.2	0.4	0.3	0.3	0.9	0.6	0.7	0.7	0.8	0.9	0.2		
	PrecStd	3.8	0.6	0.6	0.5	0.6	0.7	0.8	1.7	1.7	1.2	1.0	0.9	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.3	41.3	47.5	62.5	74.6	79.9	82.7	78.0	65.7	54.2	46.8	40.2		
		MCWB	37.5	37.5	40.2	47.3	55.6	58.8	63.7	62.5	56.9	48.9	45.3	38.3		
	2%	DB	36.5	38.4	42.8	54.5	69.1	74.7	77.8	73.6	61.9	50.5	42.7	37.6		
		MCWB	34.7	35.4	36.8	44.0	54.0	57.8	62.0	61.1	54.9	47.1	41.2	35.8		
	5%	DB	33.5	34.9	40.1	50.0	64.5	69.3	74.4	70.1	58.5	48.1	39.4	35.3		
		MCWB	32.3	32.7	35.3	40.8	51.6	55.8	60.5	59.9	52.9	45.9	38.3	34.2		
	10%	DB	31.0	32.1	37.4	46.1	59.4	65.5	70.9	66.0	56.1	46.1	36.9	33.3		
		MCWB	30.3	30.7	33.5	38.8	48.7	53.7	59.6	58.4	51.7	44.3	36.1	32.4		
	0.4%	WB	37.7	37.9	40.4	48.3	58.2	62.6	67.5	65.7	58.7	50.6	45.5	38.7		
		MCDB	40.0	40.6	46.2	60.9	69.4	74.4	76.7	73.1	62.7	52.5	46.8	39.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.9	35.6	37.7	44.6	55.2	59.4	64.5	63.0	56.2	48.1	41.5	36.0		
		MCDB	36.2	38.0	41.8	53.5	67.0	69.7	71.8	69.6	59.7	49.6	42.4	37.0		
	5%	WB	32.3	33.1	35.7	41.7	52.3	57.4	62.7	61.0	54.3	46.3	38.5	34.2		
		MCDB	33.1	34.5	39.3	48.1	63.2	66.7	71.2	67.3	57.3	47.5	39.3	35.0		
	10%	WB	30.1	30.7	33.8	39.3	49.3	55.3	60.9	59.0	52.5	44.2	36.1	32.5		
		MCDB	30.7	31.7	37.1	45.4	58.2	62.8	68.4	64.7	55.1	46.0	36.7	33.1		
	5% DB	MDBR	11.4	12.9	15.3	15.7	17.7	17.0	16.6	15.3	13.7	10.2	7.9	10.7		
		MCDBR	11.3	11.5	16.3	23.0	26.5	23.6	22.5	22.4	17.5	12.8	8.5	10.1		
		MCWBR	11.1	9.8	11.5	13.3	13.2	10.1	9.3	10.5	10.9	9.7	7.9	9.8		
		MCWBR	11.2	10.5	15.5	20.7	25.4	20.6	19.4	18.3	15.2	11.6	8.1	9.6		
(35) Mean Daily Temperature Range	5% WB	MCWBR	10.7	9.3	11.4	12.8	13.2	9.7	9.4	9.7	10.4	9.7	7.8	9.3		
	taub		0.199	0.266	0.294	0.330	0.334	0.334	0.347	0.335	0.306	0.290	0.348	0.350		
		taud	1.997	2.199	2.267	2.345	2.469	2.496	2.471	2.496	2.566	2.448	2.749	2.740		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		123	208	253	269	279	281	273	266	254	204	117	66		
		Edh at Noon	8	18	26	31	30	30	30	26	20	15	7	5		
	RadAvg		47	241	717	1229	1523	1707	1571	1191	716	292	73	19		
(45) All-Sky Solar Radiation	RadStd		5	40	70	99	180	130	171	112	85	51	15	2		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	-239	N/A	N/A	N/A	

Nomenclature: See separate page